INVESTIGATING THE SOUND ABSORPTION COEFFICIENT OF 3D-PRINTED PLA DISCS WITH VARYING PERFORATION GEOMETRIES

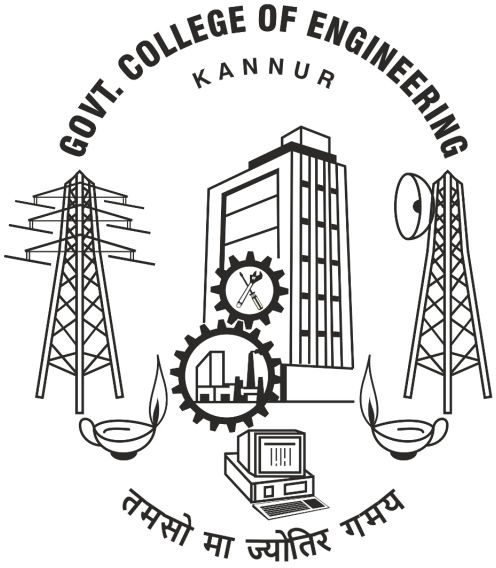
MED481 Mini Project: Report Submitted by

FATHIMA PALOT

KTU Reg. No.KNR21EC035

Towards the partial fulfilment of the requirements for the award of B. Tech. Minor in

MATERIALS AND MANUFACTURING ENGINEERING



Department of Mechanical Engineering
GOVERNMENT COLLEGE OF ENGINEERING
KANNUR - 670 563

November 2024

GOVERNMENT COLLEGE OF ENGINEERING KANNUR

DEPARTMENT OF MECHANICAL ENGINEERING

CERTIFICATE

Certified that this is a bonafide report of the mini project work done by **FATHIMA PALOT (KTU Reg. No. KNR21EC035)** on the topic **Investigating The Sound Absorption Coefficient Of 3D-printed PLA Discs With Varying Perforation Geometries** towards the partial fulfilment of the requirements for the award of the Minor Degree of Bachelor of Technology in Materials and Manufacturing Engineering from APJ Abdul Kalam Technological University during the academic year 2024-2025.

Prof. Muhammed Anees V.V.	**Dr. Sudheesh Kumar C.P.**	**Dr. Govindan P.**
Mini Project Supervisor	Mini Project Co-ordinator	Head of the Department
Department of ME	Department of ME	Department of ME
GCE Kannur	GCE Kannur	GCE Kannur

DECLARATION

I, the undersigned, hereby solemnly declare that this mini project report titled **Investigating The Sound Absorption Coefficient Of 3D-printed PLA Discs With Varying Perforation Geometries** , submitted for the partial fulfillment of the requirements for the award of the Minor degree of Bachelor of Technology in Materials and Manufacturing Engineering from APJ Abdul Kalam Technological University is a bonafide record of my own work carried out under the supervision of **Prof. Muhammed Anees V.V.**.

Wherever I have used materials (data, theoretical analysis, and text) from other sources, I have adequately and accurately cited the original sources.

I also declare that this work has not been submitted to any other institution in this University or any other University.

FATHIMA PALOT



ACKNOWLEDGMENTS

I would like to take this opportunity to extend my sincere thanks to people who helped us to make this project possible. This project will be incomplete without mentioning all the people who helped us to make it real.

I am extremely grateful to the Principal, **Dr. Rajesh K. N.**, Govt. College of Engineering, Kannur, for providing the necessary facilities. I would like to express my sincere gratitude to **Dr. Govindan P.**, Head of Department of Mechanical Engineering for giving me all the support and confidence to work with my project. I express my sincere gratitude to our project guide, **Prof. Muhammed Anees V. V.** and project co-ordinator, **Dr. Sudheesh Kumar C. P.** for their great support given during this project.

I would like to thank all the teaching and non-teaching staff of Govt. College of Engineering, Kannur for their valuable help in the successful completion of my project. I extend my thanks to family members and especially to my friends for encouraging and helping me in critical situations and make the project successful.

ABSTRACT

This project investigates the Sound Absorption Coefficient (SAC) of a disc-shaped object made from Polylactic Acid (PLA), a sustainable, biodegradable, and recyclable polymer widely used in 3D printing. PLA is known for its environmental friendliness, making it a suitable choice for developing acoustic materials. In this study, 3D-printed PLA discs with various perforation geometries is examined to understand its sound absorption capabilities, particularly for applications requiring lightweight and eco-friendly materials.The experimental investigation involves sound absorption measurements conducted in an impedance tube, a widely recognized tool for acoustic testing. Circular, square, and triangular perforations are created in the disc, each affecting the material's acoustic performance in different ways. The findings have potential applications in sustainable building materials, automotive sound insulation, and personal audio environments, offering an eco-friendly alternative to conventional sound-absorbing materials.

TABLE OF CONTENTS

ABSTRACT iii

LIST OF FIGURES vi

LIST OF TABLES vii

LIST OF ABBREVIATIONS vii

CHAPTER 1. INTRODUCTION 1

1.1 Objectives 2

1.2 Motivation 2

CHAPTER 2. LITERATURE SURVEY 3

CHAPTER 3. METHODOLOGY 6

3.1 Modeling and Fabrication Process for Perforated PLA Discs 6

3.1.1 Perforation Dimensions for Consistent Volume Removal 6

3.1.1.1 Perforation Dimensions for 100 mm Diameter Disc . . . 6

3.1.1.2 Perforation Dimensions for 29 mm Diameter Disc 7

3.1.2 3D Modeling of PLA Discs in Fusion 360 8

3.1.3 3D Printing of PLA Discs 9

3.2 Measurement Process for Sound Absorption Coefficient 10

3.2.1 Experimental Setup and Equipment 10

3.2.2 Measurement Procedure 11

3.2.3 Data Analysis and Calculation of Sound Absorption Coefficient 12

CHAPTER 4. RESULTS AND DISCUSSIONS 14

4.1 Sound Absorption Results 14

4.1.1 With Circular Perforations 14

4.1.2 With Square Perforations 15

4.1.3 With Triangular Perforations 15

4.2 Comparative Analysis of Effect of Varying Perforation Geometries . . 16

CHAPTER 5. CONCLUSION **18**

REFERENCES **19**

LIST OF FIGURES

3.1 3D Models of 100 mm Diameter PLA Discs with Varying Perforations in Fusion 360 8
3.2 3D Models of 29 mm Diameter PLA Discs with Varying Perforations in Fusion 360 9
3.3 3D Printed PLA Discs 9
3.4 Schematic of the Impedance Tube for Measuring the Sound Absorption Coefficient 10
3.5 Impedance Tube 12

4.1 SAC of PLA Disc with Circular Perforations 14
4.2 SAC of PLA Disc with Square Perforations 15
4.3 SAC of PLA Disc with Triangular Perforations 16
4.4 SAC of PLA Disc with Varying Perforation Geometries 16

LIST OF TABLES

3.1 Specifications of impedance tube 11

LIST OF ABBREVIATIONS

PLA Polylactic Acid

PS Polystyrene

SAC Sound Absorption Coefficient

CHAPTER 1

INTRODUCTION

Sound Absorption Coefficient (SAC) is a measure of a material's efficiency in absorbing sound, defined as the ratio of absorbed energy to incident energy. The sound absorption coefficient of materials varies with frequency, as different frequencies affect absorption properties differently. In practical applications, sound absorption coefficients are invaluable for professionals designing industrial facilities, commercial spaces, and residential properties. This metric provides a crucial data point for evaluating materials based on sound absorption effectiveness.

Polylactic Acid (PLA) is a biodegradable thermoplastic derived from natural lactic acid sources, such as corn, maize, or milk. It offers high aesthetic appeal through its gloss and clarity, though it is naturally stiff and brittle, often requiring plasticizers for practical applications. It is especially popular in 3D printing, widely used by beginners due to its ease of use. PLA is primarily available in filament form, but can also be found as pellets or granules.

In this project, a 50% infill density is used consistently across all tests, with sound absorption measured for three different perforation shapes: circular, square, and triangular. This approach allows for direct comparisons of how each perforation geometry affects the sound absorption coefficient of PLA discs under controlled infill conditions, offering insights into optimizing acoustic performance for various applications.

1.1 Objectives

1. Measure the SAC of 3D-printed PLA discs with varying perforation geometries.
2. Examine how circular, square, and triangular perforations influence sound absorption across different frequency ranges.

1.2 Motivation

1. Industries such as construction, automotive, and consumer products are seeking eco-friendly alternatives to traditional acoustic materials.
2. Perforations play a crucial role in sound absorption, as different shapes and distributions can alter a material's ability to dissipate sound energy effectively.
3. PLA's properties, combined with the versatility of 3D printing, allow for easily adjustable perforation shapes and densities.
4. Insights from this project could guide the design of sustainable, lightweight, and customizable soundproofing solutions suitable for building construction, automotive interiors, and personal audio spaces.

CHAPTER 2

LITERATURE SURVEY

Sailesh et al. [1] presented the influence of perforations having arbitrarily varying cross-sections on the acoustic behaviour of 3D printed bio-degradable panels made of Poly Lactic Acid (PLA). Circular perforations having six different types of cross-sectional variations with two different perforation diameters are realized. Results revealed that sound absorption of perforated panels with varying cross-section is better than uniform cross-sectional perforation for the given frequency range. In the study done by Sekar et al. [2] , the acoustic panel made from polylactic acid reinforced with wood fiber composite was 3D printed by varying its thickness and infill density. The absorption performance for the thick panel can be controlled by controlling the infill density of the panel. Customizing the acoustic absorption is therefore possible for panels from biodegradable materials.

The sound absorption properties of high porosity Si3N4 ceramic foams was studied by Du et al. [3] and the effect of structural factors such as porosity, pore size and thickness of Si3N4 ceramic foams and the air gap backing on the sound absorption properties was systematically investigated. The results showed that the higher porosity and smaller pore size of Si3N4 ceramic foams would contribute to better sound absorption in almost entire frequency. In order to process an industrial product, a series of polystyrene (PS) composite foams were prepared by continuous extrusion foaming in the study conducted by Fei et al. [4]. The sound absorption range of the PS composite foams was corresponding to the cellular structure and lignin content. As a result, the maximum sound absorption coefficient at normal incidence was higher than 0.90.

Closed-cell metal foams are lightweight and durable materials resistant to high temperature and harsh conditions, but due to their fully closed porosity they are poor airborne sound absorbers. In the study put forward by Opiela et al. [5], a classic method of drilling is used for a nearly closed-cell aluminium foam to open its porous interior to the penetration of acoustic waves propagating in air, thereby increasing the wave energy dissipation inside the pores of the perforated medium. The third octave sound absorption coefficient testing was conducted by Liang et al. [6] to compare the sound absorption properties metal foam and flexible cellular materials, by using sound absorption tester with the method of transfer function. The results show that pearl wool and glass wool exhibited excellent sound absorption properties. Among of the metal foams, the foamed aluminum material exhibited optimum sound absorption properties, and is superior to flexible sound absorption materials.

Making use of the phase gradient method for diffracted wave manipulation, Fang et al. [7] proposed an acoustic metasurface with a periodic structure backed with a rigid wall for sound absorption. The results demonstrated that the designed metasurface possesses remarkable advantages in sound absorption capability compared with the individual porous elements.

A new composite structure was formed by Xie et al. [8] by filling Nomex honeycomb with polyester fibre to achieve the goal of improving its acoustic characteristics . By using the impedance tube method, the sound absorption coefficient (SAC) of porous materials was determined. The experimental results demonstrate that the increase of porosity of filler can improve the SAC of the test sample.

Lim et al. [9] studied the sound absorption of kenaf fibre specimens both under normal and random sound incidence. The normal-incidence sound absorption coefficient measurement was conducted using the impedance tube method. The effects of thickness involving full fibre and air-fibre specimen and the effect of bulk density were discussed. Thee frequency bandwidth of absorption and the level of absorption coefficient improve significantly when bulk density and thickness are increased.

For promoting the recycling of industrially discarded luffa resources and preventing environmental pollution caused by the incineration of luffa waste, abundant luffa scraps and environmentally friendly polyester fibers were used as raw materials to produce luffa fiber sound absorbing composites in the study put forward by Chen et al. [10]. Results revealed that with the increase in the thickness, the average sound absorption coefficient of the material also increased.

CHAPTER 3

METHODOLOGY

3.1 Modeling and Fabrication Process for Perforated PLA Discs

In this project, three distinct perforation geometries—circular, square, and triangular—were chosen to assess the influence of shape on the sound absorption coefficient (SAC) of 3D-printed PLA discs. The designs ensure that the volume removed by each perforation type is equivalent, allowing a consistent basis for comparing how each geometry affects sound absorption properties.

3.1.1 Perforation Dimensions for Consistent Volume Removal

For accurate comparisons, each perforation type was adjusted to remove the same volume of material from the discs, maintaining a controlled variable for geometry. The dimensions were carefully calculated to match the total volume removed in both the 100 mm diameter and 29 mm diameter discs.

3.1.1.1 Perforation Dimensions for 100 mm Diameter Disc

For the 100 mm diameter PLA disc, the thickness of the disc is **10 mm**, and the total volume removed for each perforation type was set to approximately **670 mm^3**.

- **Circular Perforations:** The circular perforations were designed with a diameter of **1 mm.** To achieve the target volume removed, **85** circular holes were distributed across the disc surface. The cumulative volume removed by these perforations is approximately **667.6 mm^3.**

- **Square Perforations:** For square perforations, each was designed with a side length of **1 mm.** To match the volume removed by the circular perforations, **67** square perforations were distributed evenly across the disc surface. The total volume removed by these square perforations also approximates **670 mm³.**

- **Triangular Perforations:** For the triangular perforations, an equilateral triangle with a side length of **1 mm** was used. To achieve the target volume, **155** triangular perforations were placed across the disc. The total volume removed by these triangular perforations is approximately **671.2 mm³.**

3.1.1.2 Perforation Dimensions for 29 mm Diameter Disc

For the 29 mm diameter PLA disc, the thickness of the disc is **10 mm**, and the total volume removed for each perforation type was set to approximately **280 mm³**.

- **Circular Perforations:** The circular perforations were designed with a diameter of **2 mm.** To achieve the target volume removed, **9** circular holes were distributed across the disc surface. The cumulative volume removed by these perforations is approximately **282.7 mm³.**

- **Square Perforations:** For square perforations, each was designed with a side length of **2 mm.** To match the volume removed by the circular perforations, **7** square perforations were distributed evenly across the disc surface. The total volume removed by these square perforations also approximates **280 mm³.**

- **Triangular Perforations:** For the triangular perforations, an equilateral triangle with a side length of **2 mm** was used. To achieve the target volume, **16** triangular perforations were placed across the disc. The total volume removed by these triangular perforations is approximately **277.1 mm³.**

3.1.2 3D Modeling of PLA Discs in Fusion 360

The design process for the discs with perforations was carried out in Fusion 360, following a structured approach to ensure consistent and precise modeling for each perforation geometry: circular, square, and triangular. Each perforation type was modeled on a separate disc to control for variable differences in shape alone, ensuring that the total volume removed by each type remained comparable.

The process began by creating a base model of the disc. Next, each perforation geometry was individually modeled and prepared for uniform distribution on the disc surface. To achieve a uniform pattern across the disc surface, the initial perforation shape—whether circular, square, or triangular—was selected and duplicated around the center of the disc, acting as the rotation axis. Perforations were organized in concentric rings, at uniform distances from the disc center. The number of perforations per ring varied to ensure even coverage across the surface. Once the perforations were placed, each perforation type was extruded through the entire 10 mm thickness of the disc. The resulting disc models, with consistent volume removal but varying perforation shapes, were saved as separate files.

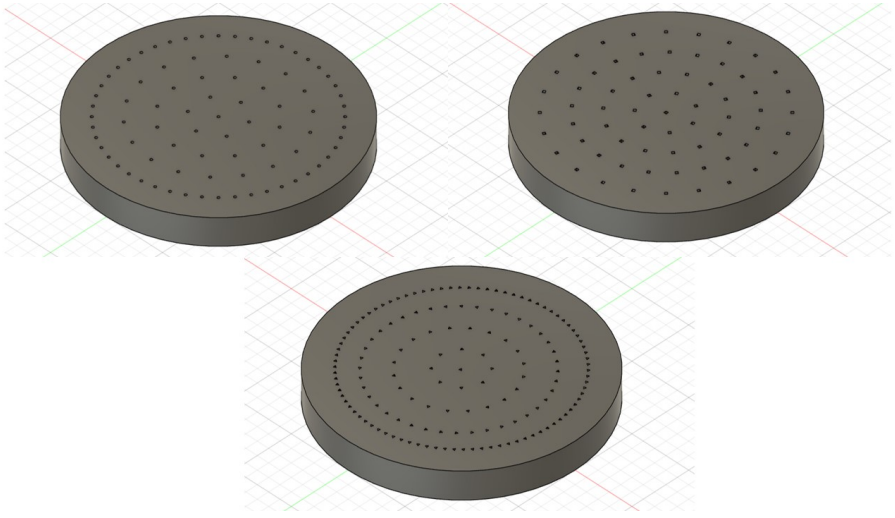


Fig. 3.1. 3D Models of 100 mm Diameter PLA Discs with Varying Perforations in Fusion 360

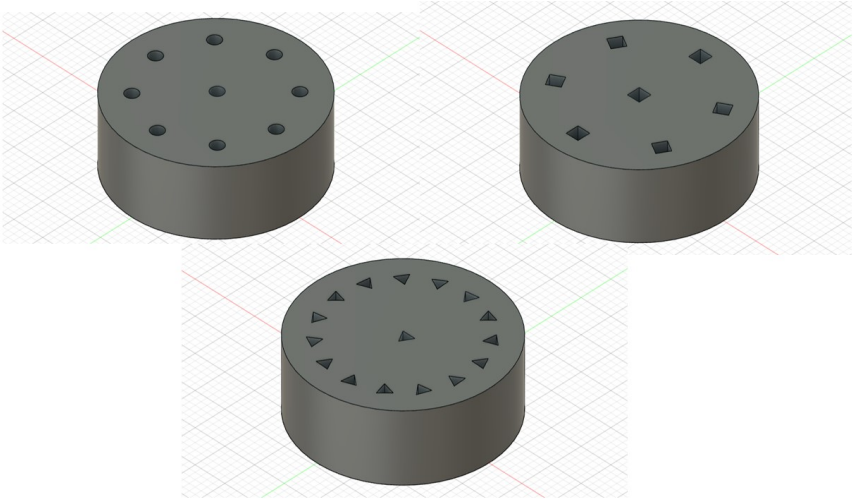


Fig. 3.2. 3D Models of 29 mm Diameter PLA Discs with Varying Perforations in Fusion 360

3.1.3 3D Printing of PLA Discs

PLA was chosen as the printing material due to its biodegradability, ease of use, and stability in print quality, making it a popular choice for acoustic material studies. Each disc was printed with precise dimensions of 100 mm and 29 mm diameters and a consistent 10 mm thickness. For each disc, the 50% infill density was specified, balancing structural integrity with a reduction in material usage and cost.

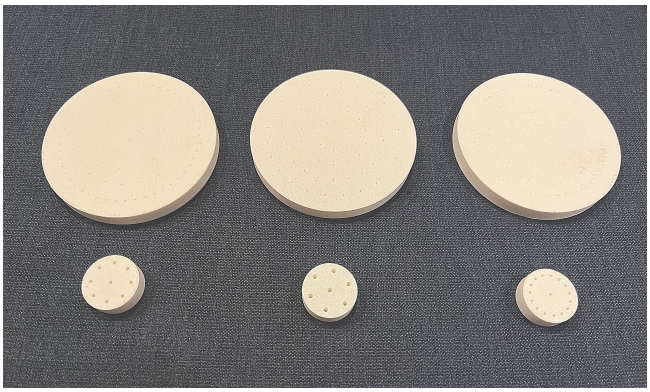


Fig. 3.3. 3D Printed PLA Discs

The circular, square, and triangular perforations, distributed evenly across each disc model, were printed with exact specifications to ensure the volume removed remained consistent across all designs. No additional post-processing was applied, as the discs' surfaces were sufficiently smooth and accurate for immediate use in

acoustic testing.

3.2 Measurement Process for Sound Absorption Coefficient

3.2.1 Experimental Setup and Equipment

To measure the sound absorption coefficient of the PLA discs, an impedance tube setup was used. The impedance tube is a standard tool for acoustic testing, designed to measure sound absorption coefficients accurately across a range of frequencies.

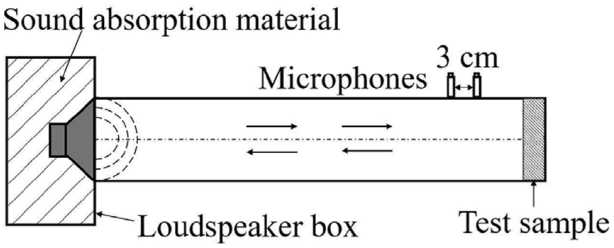


Fig. 3.4. Schematic of the Impedance Tube for Measuring the Sound Absorption Coefficient

The impedance tube with a circular cross-section having a diameter of 10 cm and a length of 90 cm is the most appropriate tube for measuring the sound absorption coefficient. The high (f_u) and low (f_l) frequencies of the tube and the distance between the two microphones (s) are calculated from the following equations:

$$f_u d < 0.5 C_o \tag{3.1}$$

$$f_l > \frac{0.75 C_o}{l - d} \tag{3.2}$$

$$f_u s < 0.45 C_o \tag{3.3}$$

where 'C_o' is the sound speed, 'l' is the overall length of the tube, and 'd' is the diameter of the tube. Also, the distance between the two microphones should be greater than 5% of the longest wavelength, which is related to the lowest interested frequency.

The impedance tube used is made of anodized aluminium and is meant to evaluate sound absorption coefficient and transmission loss according to ISO 10534-2,

Model No.	HO-AE-ITA219
Tube	Combined type (Large 100 mm & Small 29 mm)
Inner Diameter	100 mm, 29 mm
Frequency Range	100 Hz - 1600 Hz, 500 Hz - 6300 Hz
Sample Holder Inner	100 mm, 29 mm
Sample Holder Max.	200 mm, 200 mm
Standard	ISO10534-2, ASTME1050, ASTME2611
Loud Speaker	3½ “in diameter, 30 Watts, 4 Ohm
Microphone	¼” in diameter, -3dB at 1.5V sensitivity, 20-16,000 Hz frequency response
Measurement software	Holmarc Wave Analyzer 4C
Measured parameter	Sound Absorption Coefficient ()

Table 3.1. Specifications of impedance tube

ASTME1050, and ASTM E2611 test standards at normal incidence. The test sample's acoustic properties are estimated using the transfer function approach, which separates incident and reflected energy from the measured transfer function. Two high-sensitivity microphones were positioned along the tube's length to measure the incident and reflected sound waves from the disc sample.

3.2.2 Measurement Procedure

The measurement procedure was designed to obtain accurate readings of sound absorption coefficients across a specified range of frequencies. The process was conducted as follows:

1. Each 3D-printed PLA disc was prepared and inspected to ensure that the perforations were clean and free of any obstructions. The disc was then inserted into the sample holder at the end of the impedance tube, ensuring a secure fit to avoid sound leakage.

2. Before starting each measurement, the system was calibrated. The microphones were calibrated to ensure accurate response levels, and the signal generator and amplifier were set to the correct power levels.

3. The data acquisition system recorded the sound pressure levels from each microphone, capturing the amplitude and phase of incident and reflected waves

at each frequency. Each measurement was repeated multiple times to ensure consistency, and the averaged data was used in subsequent analysis.

4. This procedure was repeated for each of the six PLA disc samples with different perforation shapes and diameters. Testing each disc under identical conditions allowed for a controlled comparison of sound absorption performance based on perforation geometry.



Fig. 3.5. Impedance Tube

3.2.3 Data Analysis and Calculation of Sound Absorption Coefficient

Using the microphone data, the amplitude of the incident and reflected pressure waves was calculated at each frequency. The ratio of the reflected to incident wave amplitudes was used to derive the reflection coefficient (r), which represents the proportion of sound energy reflected by the sample.

The SAC was then calculated as:

$$\alpha = 1 - |r|^2 \tag{3.4}$$

The SAC values were plotted against frequency to produce an absorption curve for each sample, with higher values indicating a more effective sound absorption. These curves illustrate how each perforation type performs across different frequencies, highlighting specific frequency ranges where the perforation geometry signifi-

cantly enhances or reduces sound absorption. The findings were then used to compare the acoustic performance of the PLA discs based on perforation shape and disc diameter.

CHAPTER 4

RESULTS AND DISCUSSIONS

4.1 Sound Absorption Results

4.1.1 With Circular Perforations

For the PLA disc with circular perforations, in the low-frequency region (100-500 Hz), the absorption coefficient starts at approximately 16% at 100 Hz with absorption peaking around 500 Hz, with a SAC of 24%. SAC rapidly decreases to minimal levels (below 5%) between 200-500 Hz. The mid-frequency range (1000-2000 Hz) displays a gradual increase in absorption from about 5% to 15%. The most prominent peak occurs in the higher frequency range, where the absorption rises significantly to reach a maximum of approximately **63% at 4000 Hz.**

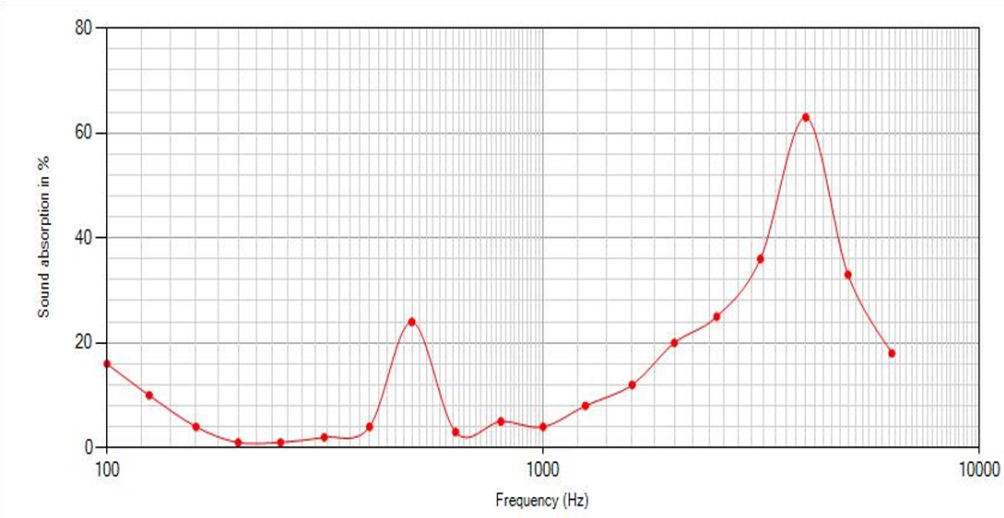


Fig. 4.1. SAC of PLA Disc with Circular Perforations

4.1.2 With Square Perforations

For the PLA disc with square perforations, in the low-frequency region (100-500 Hz), the absorption coefficient starts at approximately 18% at 100 Hz with absorption peaking around 500 Hz, with a SAC of 29%. SAC rapidly decreases to minimal levels between 200-500 Hz and then displays a gradual increase in absorption in the mid-frequency range (1000-2000 Hz) . The most prominent peak occurs in the higher frequency range, where the absorption rises significantly to reach a maximum of approximately **73% at 4000 Hz.**

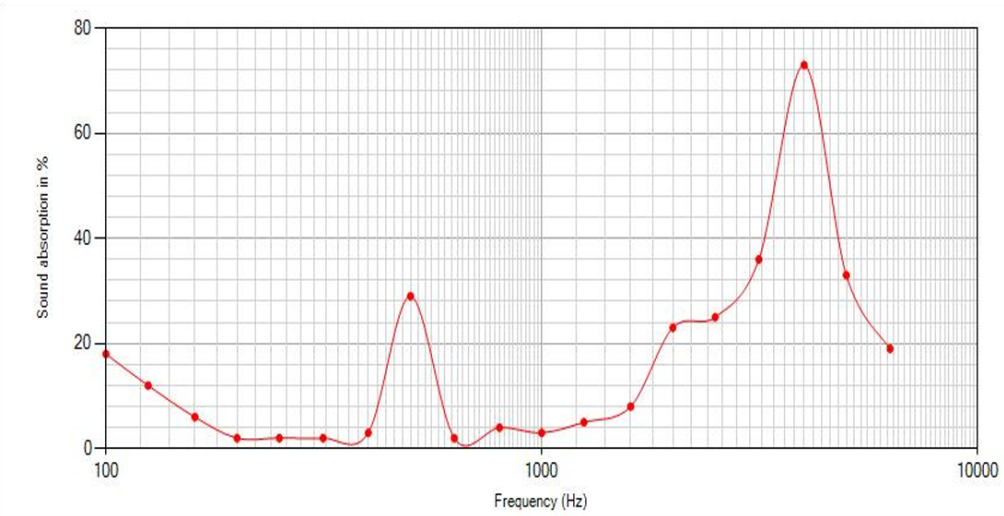


Fig. 4.2. SAC of PLA Disc with Square Perforations

4.1.3 With Triangular Perforations

For the PLA disc with triangular perforations, in the low-frequency region (100-500 Hz), the absorption coefficient starts at approximately 18% at 100 Hz with absorption peaking around 500 Hz, with a SAC of 31%. SAC rapidly decreases to minimal levels (below 5%) between 200-500 Hz and then displays a gradual increase in absorption in the mid-frequency range (1000-2000 Hz) . The most prominent peak occurs in the higher frequency range, where the absorption rises significantly to reach a maximum of approximately **98% at 5000 Hz.**

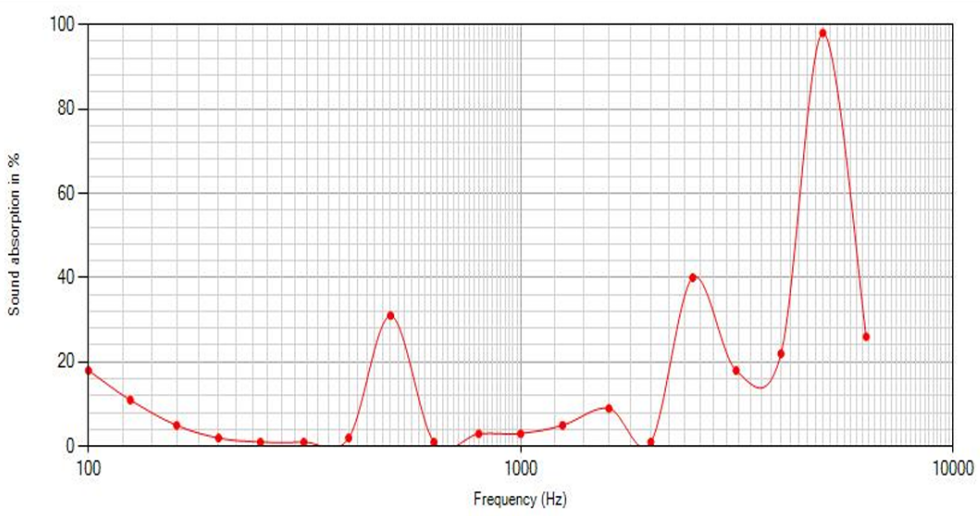


Fig. 4.3. SAC of PLA Disc with Triangular Perforations

4.2 Comparative Analysis of Effect of Varying Perforation Geometries

A comparative analysis of the sound absorption performance across circular, square, and triangular perforations highlights the influence of perforation geometry on SAC. Across both 100 mm and 29 mm disc sizes, triangular perforations consistently produced the highest SAC values, followed by square and then circular perforations.

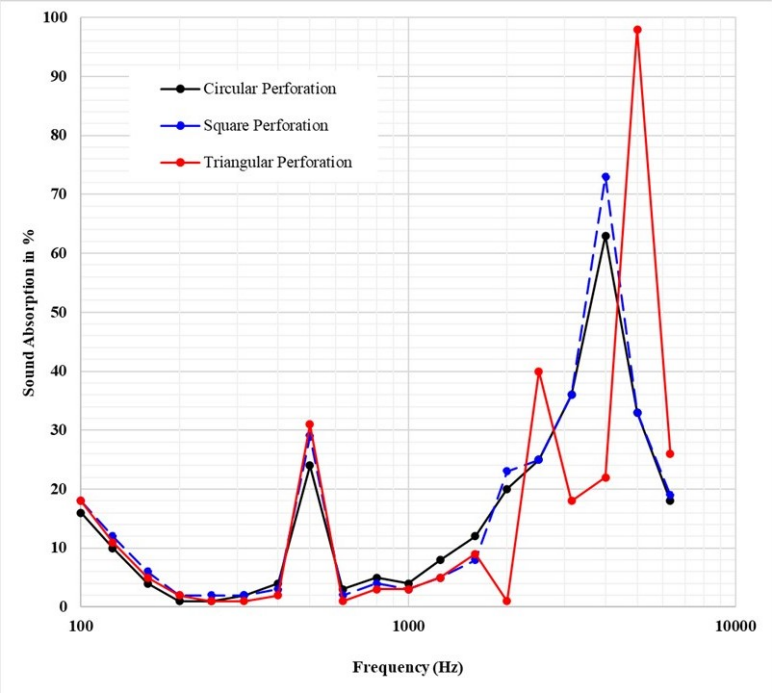


Fig. 4.4. SAC of PLA Disc with Varying Perforation Geometries

Triangular perforations exhibited the highest sound absorption at high frequencies, due to their pointed edges and sharp vertices increasing turbulence, which maximizes sound energy dissipation. Square perforations also demonstrated effective absorption, also at high frequencies, though not to the extent of triangular shapes. The sharp edges of square perforations provide enhanced sound scattering and frictional interaction with sound waves compared to circular shapes. Circular perforations produced moderate SAC values due to their smoother, rounded shape, which seems less effective at disrupting sound waves, resulting in lower absorption.

In conclusion, the comparative analysis demonstrates that perforation geometry is a critical factor in optimizing sound absorption properties of PLA discs, with triangular shapes outperforming other geometries due to their enhanced turbulence and scattering effects.

CHAPTER 5

CONCLUSION

This study investigated the sound absorption properties of 3D-printed PLA discs with circular, square, and triangular perforations, analyzing how perforation geometry and disc size affect the sound absorption coefficient (SAC). By examining discs with diameters of 100 mm and 29 mm, each with consistent volume removal and infill density, a controlled comparison was conducted across low, mid, and high frequencies.

The results indicate that triangular perforations achieved the highest SAC values across both disc sizes, particularly in the high frequency range, where sharp vertices and pointed edges facilitated increased turbulence and sound energy dissipation. Square perforations demonstrated moderate absorption performance, benefiting from sharper edges that contribute to scattering and friction, though less effectively than triangular shapes. Circular perforations provided the lowest SAC values across all frequency ranges, suggesting that smoother edges result in minimal sound wave disruption and limited absorption efficiency at higher frequencies.

These findings support the use of triangular perforations in acoustic applications where high sound absorption is essential, particularly in environments where broad frequency coverage is required. The study highlights that optimized perforation design in PLA-based materials can lead to lightweight, cost-effective, and environmentally friendly solutions for sound absorption, with practical applications in noise control across various industries.

REFERENCES

[1] R. Sailesh, L. Yuvaraj, J. Pitchaimani, M. Doddamani, and L. Chinnapandi, "Acoustic behaviour of 3d printed bio-degradable micro-perforated panels with varying perforation cross-sections," *Applied Acoustics*, vol. 174, p. 107769, 2021.

[2] V. Sekar, S. Eh Noum, S. Sivanesan, A. Putra, D. Chin Vui Sheng, and D. Kassim, "Effect of thickness and infill density on acoustic performance of 3d printed panels made of natural fiber reinforced composites," *Journal of Natural Fibers*, vol. 19, no. 13, pp. 7132–7140, 2022.

[3] Z. Du, D. Yao, Y. Xia, K. Zuo, J. Yin, H. Liang, and Y. Zeng, "The sound absorption properties of highly porous silicon nitride ceramic foams," *Journal of Alloys and Compounds*, vol. 820, pp. 107–114, 2020.

[4] Y. Fei, W. Fang, M. Zhong, J. Jin, P. Fan, J. Yang, Z. Fei, L. Xu, and F. Chen, "Extrusion foaming of lightweight polystyrene composite foams with controllable cellular structure for sound absorption application," *Polymers*, vol. 11, no. 1, p. 106, 2019.

[5] K. Opiela, T. Zieliński, T. Dvorák, and S. Kúdela Jr, "Perforated closed-cell aluminium foam for acoustic absorption," *Applied Acoustics*, vol. 174, p. 107706, 2021.

[6] L. Liang, X. Wu, N. Ma, J. Du, and M. Liu, "The sound absorption properties comparison of metal foams and flexible cellular materials," *Materials Science Forum*, vol. 933, pp. 357–366, 2018.

[7] Y. Fang, X. Zhang, and J. Zhou, "Acoustic porous metasurface for excellent sound absorption based on wave manipulation," *Journal of Sound and Vibration*, vol. 434, pp. 273–283, 2022.

[8] S. Xie, S. Yang, C. Yang, and D. Wang, "Sound absorption performance of a filled honeycomb composite structure," *Applied Acoustics*, vol. 162, p. 107202,

2020.

[9] Z. Lim, A. Putra, M. Nor, and M. Yaakob, "Sound absorption performance of natural kenaf fibres," *Applied Acoustics*, vol. 130, pp. 107–114, 2018.

[10] Y. Chen, F. Yuan, Q. Su, C. Yu, K. Zhang, P. Luo, D. Hu, and Y. Guo, "A novel sound absorbing material comprising discarded luffa scraps and polyester fibers," *Journal of Cleaner Production*, vol. 245, p. 118917, 2020.